Freight Rate Prediction Assessment

Eduardo Ander Quintero Sánchez

1 Problem and data

The task is a tabular regression problem. The objective is to predict the freight load rate, using `posted_rate` as the training target and producing `predicted_rate` for the final outputs.

The labeled `train-test.csv` contains 48,000 loads from January 1 to October 31, 2025. The final validation set contains 12,000 loads from November 1 to December 31, 2025.

There is also a fixed December input used for the required chart. In this set the pickup, delivery, distance, equipment, and weight stay fixed, and only the date changes from December 1 to December 31.

Because the final data comes after the labeled data in time, I treated the time ordering as an important part of the validation design.

2 Data exploration and quality

The first exploration showed that `distance` is the clearest individual predictor of the target. Its linear correlation with `posted_rate` is approximately 0.91.

The target has a skewed distribution and some large values. I also checked a `log1p` transformation later during modeling.

Some other observations affected the preprocessing and feature decisions:

- `weight` and `market_index` contain missing values.
- Validation contains 8 pickup cities and 8 delivery cities that do not appear in the labeled training data.
- Pickup and delivery city coordinates are almost redundant with the city labels in this dataset, and the provided distance is also very close to a Haversine distance calculation.
- The December input has fewer columns than the train and validation data. In particular, it does not include coordinates or `market_index`.
- Daily rates show a lot of noise, while monthly medians are more stable.

During a final data-quality check I also found invalid negative `weight` values: 292 rows in the labeled data and 145 rows in the validation data.

Negative freight weight is not valid. I cannot know if these values are only sign errors, so I did not assume that taking the absolute value was correct. I marked them as missing and let the same numeric median imputation used by the model handle them.

3 Validation and preprocessing

I used the last two months of the labeled data, September and October, as a temporal holdout.

I prefer this over a random split because the real validation set comes after the training period. A random split would mix early and late dates and would be less similar to the final prediction situation.

For preprocessing:

- numerical missing values are filled with the training median;
- categorical missing values are filled with the most frequent value;
- pickup, delivery, and equipment are one-hot encoded;
- unknown categories are ignored during transformation, so unseen validation cities do not break the pipeline;
- the date is represented with `month` and `dayofyear`.

The final feature set used by the delivery model is:

```
distance, weight, equipment, pickup, delivery, month, dayofyear, market_index
```

In the linear feature checks, adding coordinates did not improve the holdout result, and `quote_signal` also added little. Among the extra columns, `market_index` produced most of the improvement.

For the fixed December input, `market_index` is not available. I leave it as missing and the same fitted median imputation handles it.

4 Model comparison

I first used the mean training target as a simple baseline and then compared a linear regression pipeline with a HistGradientBoostingRegressor (HGB).

The main holdout MAE results after the final data-quality correction were:

Model	**Holdout MAE ($)**
Training mean baseline	1178.80
Linear regression	142.36
HGB, log target, default parameters	128.42
HGB, log target, small parameter search	123.60

The HGB model is better on the temporal holdout. A small parameter search improves the MAE to approximately $124.

However, there is another constraint in the assessment: the same solution also needs to generate the fixed December predictions.

When I fit HGB on all labeled data and predict December, the predictions stay almost flat. The training data ends in October and tree-based models do not extrapolate a continuous time trend beyond the range they have seen. I also tried a continuous day index, but the December result stayed flat.

The linear model is slightly worse on the temporal holdout, but it preserves a date-dependent extrapolation outside the training period.

Because I need one model that works consistently for both the 12,000 validation loads and the fixed December output, I use the linear regression pipeline as the final delivery model.

5 December result

The final model is refit using all available labeled data and then used for both required outputs.

For the fixed December load, the model predicts a gradual increase from about \$834 at the beginning of the month to about \$922 at the end.

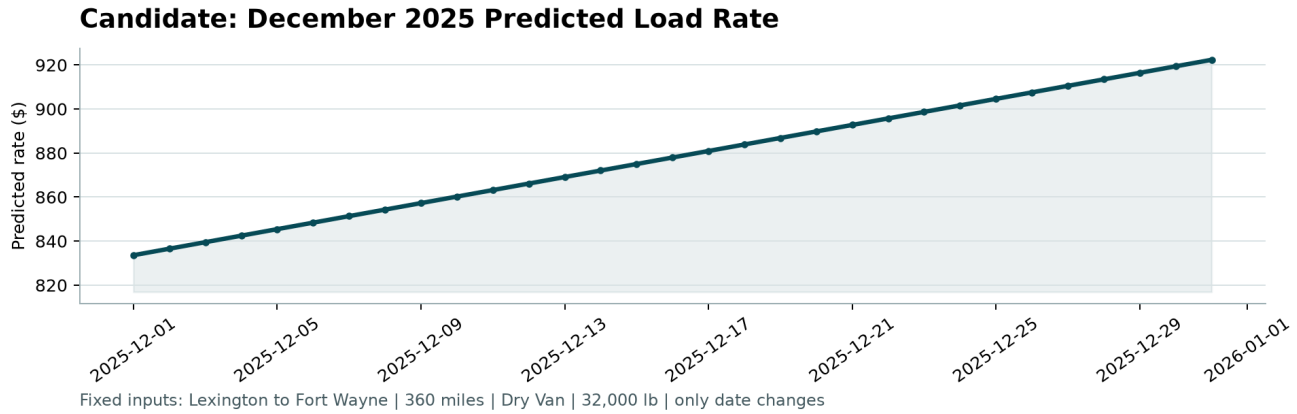


Figure 1: Fixed December 2025 prediction generated with the provided scorer.

This chart should not be interpreted as a detailed forecast of daily freight market behavior.

All load characteristics are fixed in this experiment and only the date changes. The line is therefore mainly an extrapolation of the temporal effect learned by the linear model.

The more flexible HGB model gives a better holdout MAE, but it cannot extrapolate this time component outside the training range in the same way. This is the main trade-off behind the final model choice.